WASP-43b

R-0480

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_231203` · created 2026-07-20T05:53:20+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460281.9934 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.027
Transit depth (Rp/Rs)^2	3.91 +/- 1.05 [%]
Orbital Inclination (inc)	79.2 +/- 1.43
Airmass coefficient 1 (a1)	1.109 +/- 0.013
Airmass coefficient 2 (a2)	-0.071 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	None
Optimal Aperture	3.6
Optimal Annulus	13.04
Transit Duration (day)	0.0416 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.027 vs 0.1594 (deviation 1.43σ)
Duration fit vs published	0.0416 d vs 0.0512 d (deviation 1.14σ)
Mid-time O–C	-6.1 min
Depth SNR	7.3
Residual scatter	1.22 %

Light curve

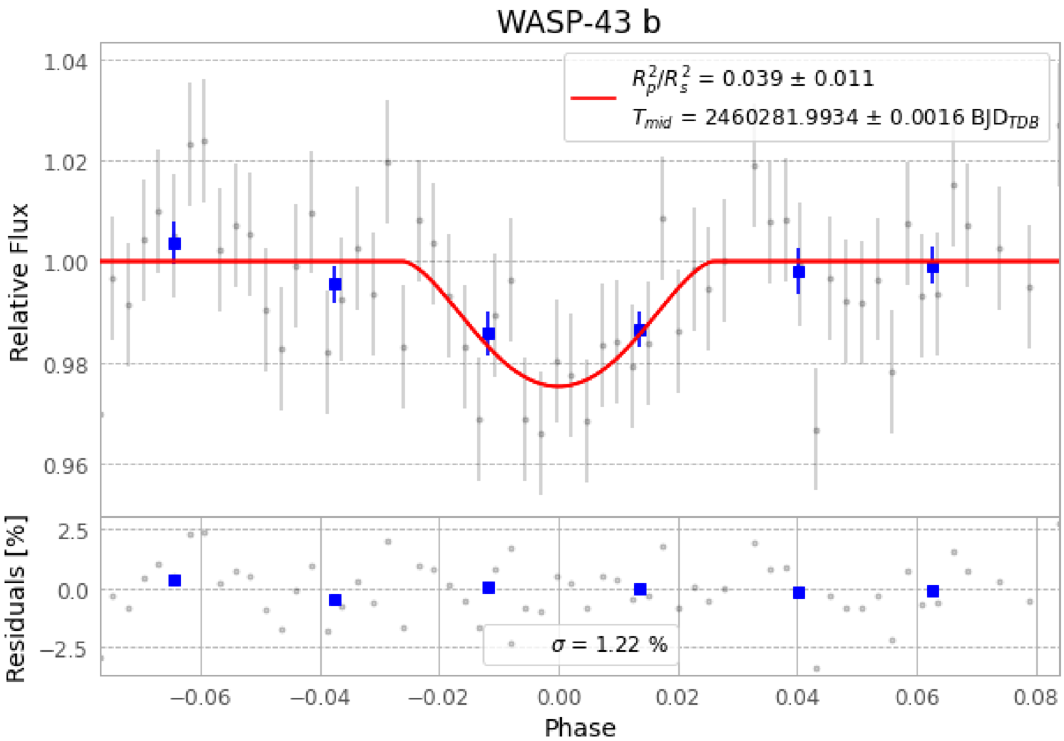

Figure 1 — FinalLightCurve_WASP-43 b_2023-12-03.png (SHA-256 8c2875c68f836bdb…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 251], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 de0d6b75bf897f0c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

62 FITS frames (41 MB), `WASP-43231203101846.FITS` → `WASP-43231203133315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-231203.log (e57361fd2359…), FinalLightCurve_WASP-43 b_2023-12-03.png (8c2875c68f83…), FinalLightCurve_WASP-43 b_2023-12-03.csv (da8893a26c65…)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)